

Build the geometry in; don't hope it emerges

Reflection symmetry is either hoped for (left) or imposed by construction (right). Four experiments, real measured numbers.

EMERGENT

Hope it emerges from self-supervised training

E1 — standard SSL pretraining

FAILS

Symmetry does not emerge

Mirror slope -0.70 (no flip; -1 = ideal)

Decodability 0.198 [0.175,0.221]

no better than floor 0.245 [0.214,0.275]

E3b — reflection-augmented SSL

FAILS

Augmentation buys no equivariance

Free-readout slope -0.51 (was -0.70), but the
untrained floor is -0.82 → confounded

Signed axis 0.062 [0.011,0.114] fails to clear
untrained floor 0.078 [0.013,0.144]

BUILT-IN

Guaranteed by construction

E2 — equivariant readout head

SUCCEEDS

Succeeds — exact and better

Mirror slope -1.0000 exact (any weights)

Decodability 0.273 [0.253,0.294]

beats free readout 0.198 [0.175,0.221]

E3a — equivariant encoder (frame avg)

SUCCEEDS

Succeeds — exact by construction

Token-equivariance error = 0.0
exact, with zero retraining

**The geometry must be imposed by construction - it does not emerge from self-supervision,
with or without reflection augmentation.**

E3b is a single-seed, Stage-0 normal-only prototype - treat as indicative, not confirmatory.

All results are transductive (internal validity only; the encoder was trained on the evaluated sequences).

Source video is the independent unit; folder labels are dataset annotations, not diagnoses.